# Teams and Groups as Nested Systems: The Collective Markov Blanket

## Executive Summary

Teams are not just collections of individuals — they are **systems in their own right**, with their own Markov blankets, internal states, and adaptive behaviors. Under Active Inference, a team is a nested inference agent: simultaneously a component of the larger organization and an autonomous whole with its own boundary, sensing apparatus, and action capacity. This module examines how team boundaries form, what makes a team function as a coherent system versus a loose collection, and how multi-scale organization enables collective intelligence.

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## Learning Objectives

1. Explain how a **team** constitutes a system with its own Markov blanket
2. Analyze **boundary permeability** — how open or closed the team is to external information and influence
3. Understand **multi-scale organization** — how individuals nest within teams, teams within departments
4. Identify when a team is **coherent** (functioning as a unified system) versus **fragmented** (a loose collection)
5. Apply the concept of **boundary management** to team design and collaboration

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## Key Concepts

### 1. The Team as a System

A team becomes a system when its members develop:

- **Shared internal states**: Common knowledge, shared mental models, joint understanding of the task

- **A team boundary (Markov blanket)**: Distinguishing team members from non-members, regulating information flow in and out
- **Collective sensing**: The team perceives its environment through combined individual observations
- **Collective action**: The team acts on its environment as a coordinated unit, not just as individuals acting in parallel

  **When is a group NOT a team?** When individuals share a manager but have no shared mental model, no collective sensing, and no coordinated action — they are a group, not a system.

## 2. Boundary Permeability

Team boundaries vary in their **permeability** — how easily information, people, and influence cross the boundary:

| Permeability | Advantages | Risks | Example |
|---|---|---|---|
| **Low (closed)** | Strong identity, deep trust, rapid internal coordination | Groupthink, information isolation, stale assumptions | Elite military units, founding startup teams |
| **High (open)** | Fresh perspectives, rapid adaptation, broad information access | Weak identity, coordination overhead, knowledge leakage | Cross-functional task forces, open-source teams |
| **Adaptive** | Adjusts permeability based on context — tightens during execution, opens during sensing | Requires sophisticated boundary management | Agile teams in well-run organizations |

**Case Study — Pixar's Braintrust**: Pixar's Braintrust is a peer review meeting where directors present in-progress films for candid feedback from other directors and creatives. The system works because Braintrust meetings temporarily raise boundary permeability — opening teams to external critique — while maintaining strong team identity. Critically, the Braintrust has no authority: feedback is advisory, preserving the team's autonomy in responding.

## 3. Multi-Scale Organization

Collective intelligence emerges from the **nesting** of systems at multiple scales:

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Organization
├── Division (inference about divisional strategy)
│   ├── Team A (inference about project delivery)
│   │   ├── Individual a1 (inference about task execution)
│   │   └── Individual a2
│   └── Team B (inference about customer success)
│       ├── Individual b1
│       └── Individual b2
└── Division (inference about market position)
    └── ...
```

Each level:

- Maintains its own generative model at the appropriate scale
- Receives "summarized" sensory states from the level below (e.g., team aggregates individual observations)
- Passes "contextualized" priors down to the level below (e.g., team receives strategic direction from division)

### 4. Team Coherence: What Makes a Team a System?

| Dimension | Coherent Team | Fragmented Group |
|---|---|---|
| **Shared model** | Members have aligned understanding of the goal and environment | Each person has a private mental model |
| **Boundary** | Clear sense of who is "in" and who is "out" | Membership is ambiguous |
| **Joint attention** | Team members attend to the same signals | Each person attends to different things |
| **Coordinated action** | Actions are planned and executed jointly | Actions are parallel but uncoordinated |
| **Collective learning** | The team updates its shared model together | Learning is individual and siloed |

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## Applications

### Team System Health Check

Rate your team on coherence dimensions (1 = fragmented, 5 = coherent):

| Dimension | Score (1–5) | Evidence |
|---|---|---|
| Shared mental model | | |
| Clear boundary | | |
| Joint attention | | |
| Coordinated action | | |
| Collective learning | | |

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## Cross-References

- For organizations as bounded systems, see Organizational Systems: Systems
- For competitive ecosystem boundaries, see Strategic Modeling: Systems
- For digital platform boundaries, see Digital Transformation: Systems

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## Summary

| Concept | Team/Collective Meaning |
|---|---|
| System | A team with its own boundary, states, and adaptive behavior |
| Markov blanket | What information flows into and out of the team |
| Boundary permeability | How open or closed the team is to external influence |
| Multi-scale organization | Individuals within teams within departments |
| Team coherence | Whether the group functions as a unified inference agent |

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## References


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